

Analyzing Determinants of Poor Child Health in India: A Data-Driven Study using NFHS-5 Dataset

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Code link: `GitHub: Analysis_of_NFHS - 5_data_for_Child_Health_Factors`

Abstract—This study leverages data from the National Family Health Survey (NFHS-5) to investigate the key socio-demographic and health-related determinants responsible for poor child health outcomes in Indian districts. Using machine learning models such as Random Forest and Ridge Regression, along with feature selection techniques like RFECV, Variance Threshold, and correlation analysis, we identify the top contributing factors to conditions such as anemia, underweight, stunting, and wasting among children below the age of 5. This project aims to help stakeholders and policymakers prioritize interventions and health schemes.

Index Terms—Child Health, NFHS-5, India, Feature Selection, Ridge Regression, Random Forest, Anemia, Stunting, Underweight, Data Science

I. INTRODUCTION

The National Family Health Survey (NFHS-5) provides vital data regarding health and family welfare in India. This project utilizes district-level data from NFHS-5 to explore the underlying factors behind poor health indicators in children below the age of 5 and infants below 6 months. Our objective is to identify strong predictors among maternal health, education, sanitation, and family planning metrics that contribute significantly to malnutrition indicators like anemia, underweight, wasting, and stunting.

II. DATA AND METHODOLOGY

A. Dataset Description

The dataset used in this study consists of two main components:

- **Child Health Data:** Includes district-wise indicators like percentage of children who are underweight, anemic, stunted, or wasted.
- **Female and Maternal Factors:** Comprises features such as BMI, education, menstrual hygiene, sanitation, access to healthcare, and contraceptive usage among women.

B. Preprocessing and Feature Selection

Although, using feature reduction methods like correlation-based elimination or variance thresholding may not be ideal in this study, as these techniques could inadvertently discard important features that are highly relevant to the target variable, such as **Anaemic Women**. Although some features might show high correlation or low variance, they could still provide

crucial insights into the underlying relationships between maternal health and child health outcomes. We still used the following feature reduction techniques for demonstration purpose:

- **Variance Threshold:** Eliminated low-variance features that add little information.
- **Correlation Matrix:** Removed highly correlated features to reduce redundancy.
- **RFECV (Recursive Feature Elimination with Cross-Validation):** Identified the most significant predictors using Ridge regression.
- **We used Coefficients generated by Ridge Regression to identify which of the factors are negatively correlated with the target.**

C. Modeling Techniques

We applied Random Forest and Ridge Regression models to evaluate feature importance and generate predictions for each target health outcome. The models were evaluated using R^2 Score and Mean Squared Error.

III. RESULTS AND KEY FINDINGS

A. Anemia among Children

- **R^2 Score:** 0.67
- **Top Predictors:**
 - Women age 15-19 years who are anemic
 - Female Schooling
 - Unmet Spacing and Unmet Family Planning Needs

B. Underweight Children

- **R^2 Score:** 0.69
- **Top Predictors:**
 - Low BMI in Mothers (strongest factor)
 - Menstrual Hygiene (negatively correlated)
 - Child Marriage
 - Overweight Women (inverse relation)

C. Wasted Children

- **R^2 Score:** $\approx$ 0.5 (Low predictive strength)
- **Key Indicators:**
 - Low BMI in Mothers
 - Overweight Women (inverse relation)

D. Stunted Children

- **Top Predictors:**

- Menstrual Hygiene (negatively correlated)
- Low BMI in Mothers
- Women Literacy (negatively correlated)

IV. DISCUSSION

Feature reduction methods like correlation and variance threshold were explored but not solely relied upon because they risked eliminating important predictors such as percentage of anemic women, which may have low variance but are strong explanatory variables. Hence, we employed more robust and interpretable feature selection methods like RFECV to retain meaningful attributes.

V. CONCLUSION

This study highlighted critical maternal and social factors influencing child health in India. Factors like Low BMI, anemia in adolescent females, menstrual hygiene, child marriage, and women literacy emerged as significant predictors. These insights can aid in better policy-making and targeting of public health interventions in high-risk districts.

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VI. DATA ACCESS

The open access raw NHFS-5 Data was accessed from website "Data.Gov.in". Link to the data source : <https://www.data.gov.in/catalog/national-family-health-survey-nfhs-5>